

When we have equal sample sizes, this formula simplifies to

$$SSC = \frac{n(\hat{l})^2}{\sum_{i=1}^t a_i^2}$$

Associated with each such sum of squares is 1 degree of freedom. Thus, we can obtain $t - 1$ orthogonal contrasts such that the sum of squares treatment, which has $t - 1$ degrees of freedom, equals the total of the $t - 1$ sum of squares associated with each of the contrasts. The following example illustrates these calculations.

EXAMPLE 9.3

Various agents are used to control weeds in crops. Of particular concern is the overusage of chemical agents. Although effective in controlling weeds, these agents may also drain into the underground water system and cause health problems. Thus, several new biological weed agents have been proposed to eliminate the contamination problem present in chemical agents. Researchers conducted a study of biological agents to assess their effectiveness in comparison to the chemical weed agents. The study consisted of a control (no agent), two biological agents (Bio1 and Bio2), and two chemical agents (Chm1 and Chm2). Thirty 1-acre plots of land were planted with hay. Six plots were randomly assigned to receive one of the five treatments. The hay was harvested and the total yield in tons per acre was recorded. The data are given here.

Agent	1	2	3	4	5
Type	None	Bio1	Bio2	Chm1	Chm2
$\bar{y}_{i\cdot}$	1.175	1.293	1.328	1.415	1.500
s_i	.1204	.1269	.1196	.1249	.1265
n_i	6	6	6	6	6

Determine four orthogonal contrasts and demonstrate that the total of the four sums of squares associated with the four contrasts equals the sum of squares for treatment (between-samples).

Solution An analysis of variance was conducted on these data yielding the results summarized in the AOV table given here.

Source	df	SS	MS	F	p-value
Treatment	4	.3648	.0912	5.96	.0016
Error	25	.3825	.0153		
Totals	29	.7472			

From the AOV table, we have that $SS_{\text{Trt}} = .3648$. We will now construct four orthogonal contrasts in the five treatment means and demonstrate that SS_{Trt} can be partitioned into four terms, each representing a 1-degree of freedom sum of square associated with a particular contrast. Table 9.3 contains the coefficients and sum of squares for each of the four contrasts.

TABLE 9.3 Sum of squares computations for weed control experiment

Contrast	Treatment					$\sum_{i=1}^5 a_i^2$	$\hat{l}$	SSC _i
	1(Cntrl)	2(Bio1)	3(Bio2)	4(Chm1)	5(Chm2)			
	a_1	a_2	a_3	a_4	a_5			
Control vs. Agents	4	-1	-1	-1	-1	20	-.836	.2097
Biological vs. Chemical	0	1	1	-1	-1	4	-.294	.1297
Biol vs. Bio2	0	1	-1	0	0	2	-.035	.0037
Chm1 vs. Chm2	0	0	0	1	-1	2	-.085	.0217
$\bar{y}_i$	1.175	1.293	1.328	1.415	1.500			.3648

To illustrate the calculations involved in Table 9.3, we will compute the sum of squares associated with the first contrast, control vs. agents. First, note that the contrast represents a comparison of the yield for the control treatment versus the average yield of the four active agents. We initially would have written this contrast as

$$\begin{aligned}
 l &= \mu_1 - \frac{(\mu_2 + \mu_3 + \mu_4 + \mu_5)}{4} \\
 &= (1)\mu_1 + \left(\frac{-1}{4}\right)\mu_2 + \left(\frac{-1}{4}\right)\mu_3 + \left(\frac{-1}{4}\right)\mu_4 + \left(\frac{-1}{4}\right)\mu_5
 \end{aligned}$$

However, we can multiply each coefficient by 4 and change the coefficients from

$$a_1 = 1 \quad a_2 = \frac{-1}{4} \quad a_3 = \frac{-1}{4} \quad a_4 = \frac{-1}{4} \quad a_5 = \frac{-1}{4}$$

to

$$a_1 = 4 \quad a_2 = -1 \quad a_3 = -1 \quad a_4 = -1 \quad a_5 = -1$$

Next, we calculate

$$\sum_{i=1}^5 a_i^2 = (4)^2 + (-1)^2 + (-1)^2 + (-1)^2 + (-1)^2 = 20$$

and

$$\begin{aligned}
 \hat{l} &= (4)(1.175) + (-1)(1.293) + (-1)(1.328) + (-1)(1.415) + (-1)(1.500) \\
 &= -.836
 \end{aligned}$$

Finally, we can obtain the sum of squares associated with the contrast from

$$SSC_1 = \frac{(\hat{l})^2}{\sum_{i=1}^5 (a_i^2/n_i)} = \frac{n(\hat{l})^2}{\sum_{i=1}^5 a_i^2} = \frac{6(-.836)^2}{20} = .2097$$

The remaining three sums of squares are calculated in a similar fashion. From Table 9.3 we thus obtain

$$SSC_1 + SSC_2 + SSC_3 + SSC_4 = .2097 + .1297 + .0037 + .0217 = .3648 = SS_{\text{Tt}}$$

You will verify that the four contrasts are indeed mutually orthogonal in an exercise.